1.2.4 Equivalent rational numbers

We know how to write equivalent fractions when a fraction is given. Since a rational number can be represented by a fraction, we can think of equivalent rational numbers, duly obtained through equivalent fractions.

Suppose a rational number is in fractional form. Multiply its numerator and denominator by the same non-zero integer to obtain a rational number which is equivalent to it.

For example,

$-\frac{2}{3}$ is equivalent to $\frac{-6}{9}$ since $\frac{-2}{3}=\frac{-2\times 3}{3\times 3}=\frac{-6}{9}$

$-\frac{2}{3}$ is also equivalent to $\frac{10}{-15}$ since $\frac{2}{-3}=\frac{2\times 5}{-3\times 5}=\frac{10}{-15}$

Thus, $-\frac{2}{3}=\frac{-6}{9}=\frac{10}{-15}$.



Try these

1. $\frac{7}{3}=\frac{?}{9}=\frac{49}{?}=\frac{-21}{?}$
2. $\frac{-2}{5}=\frac{?}{10}=\frac{6}{?}=\frac{-8}{?}$

1.2.5 Rational numbers in Standard form

Observe the following rational numbers: $\frac{4}{5},\frac{-3}{7},\frac{1}{6},\frac{-4}{13},\frac{-50}{51}$. Here, we see that

i) the denominators of these rational numbers are all positive integers

ii) 1 is the only common factor between the numerator and the denominator of each of them and

iii) the negative sign occurs only in the numerator.

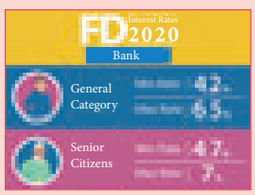
Such rational numbers are said to be in standard form.

A rational number is said to be in standard form, if its denominator is a positive integer and both the numerator and denominator have no common factor other than 1.

If a rational number is not in the standard form, then it can be simplified to arrive at the standard form.



The collection of rational numbers is denoted by the letter ℚ because it is formed by considering all quotients, except those involving division by 0. Decimal numbers can be put in quotient form and hence they are also rational numbers.



Example 1.2

Reduce to the standard form: (i) $\frac{48}{-84}$ (ii) $\frac{-18}{-42}$

Solution:

(i) **Method 1:**

$$\frac{48}{-84}=\frac{48\div(-2)}{-84\div(-2)}=\frac{-24\div 2}{42\div 2}=\frac{-12\div 3}{21\div 3}=\frac{-4}{7}$$ (dividing by –2, 2 and 3 successively)

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